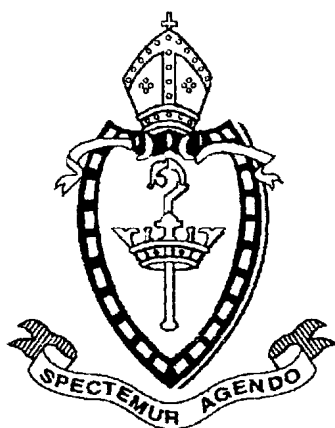


NEWCASTLE GRAMMAR SCHOOL



YEAR 12 2003 MATHEMATICS TRIAL EXAMINATION

*Time allowed – Three hours
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- ALL questions are of equal value.
- All necessary working should be shown in every question.
Marks may be deducted for careless or badly arranged work.
- Standard integrals are printed on page 10.
- Board-approved calculators may be used.
- Answer each question in a SEPARATE Writing Booklet.
- You may ask for extra Writing Booklets if you need them.

QUESTION 1 Use a SEPARATE Writing Booklet.**Marks**

- a) Calculate the value of $\frac{\sqrt{4\pi}}{3 \cdot 6^2 - 9 \cdot 8}$ correct to four significant figures 2
- b) Express $\frac{6}{\sqrt{3}-1}$ with a rational denominator 2
- c) Differentiate $6 - x^3$ 2
- d) Solve $\frac{x}{2} + \frac{x}{3} = 1$ 2
- e) Integrate $\frac{4}{x}$ 2
- f) Factorise completely $9 - 16t^2$ 2

QUESTION 2 Use a SEPARATE Writing Booklet.

- a) Differentiate:
- i) $y = e^{\sin x} + \frac{x^4}{2}$ 3
- ii) $y = \frac{\log_e x}{x}$ 3
- b) Sketch the graph with the equation $y = x - x^2$ showing all intercepts: 2
- c) Solve $|x + 4| = 1$ 2
- d) Give the exact value for $\sec 210^\circ$ 2

QUESTION 3 Use a SEPARATE Writing Booklet.**Marks**

- a) The first term of an arithmetic sequence is 6 and the common difference is 9. **3**
- i) Write down the expression for the n^{th} term
- ii) Which term of this sequence is 4623 ?
- b) Consider the points O (0, 0), A (-1, 3) and B (11, -6)
- i) Find the gradient of line AB **1**
- ii) Show that the equation of AB is $3x + 4y - 9 = 0$ **2**
- iii) Find the equation of line L, which passes through O and is parallel to line AB **2**
- iv) The point P, (4, k), lies on line L. Find the value of k **2**
- v) Calculate the perpendicular distance from P to AB **2**

QUESTION 4 Use a SEPARATE Writing Booklet.**Marks**

a) Find

i) $\int \cos 2x \, dx$ 2

ii) $\int \frac{dx}{2x+3}$ 2

iii) $\int e^{3x} \, dx$ 2

- b) Bank X pays compound interest, compounded annually. 3
Bank Y pays simple interest. \$5000 is invested in Bank X
and also in Bank Y at 9% p.a. for 6 years - at both banks.
Find the difference between the compound interest and
simple interest earned at each bank.

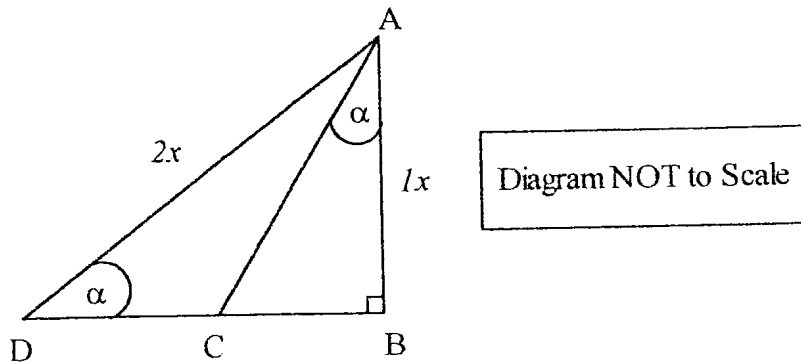
- c) For what values of k is $(3-k)x^2 + (3-k)x + 1$ positive definite? 3

QUESTION 5 Use a SEPARATE Writing Booklet.

Marks

- a) In the diagram below, $AD = 2 \times AB$ and $\angle ADC = \angle BAC$

5



- i) By writing an expression for $\sin \alpha$, show that $\alpha = 30^\circ$
- ii) Hence find the size of $\angle DAC$
- iii) If $DC = 2 \text{ cm}$ find the length of AB

- b) Solve $9^x + 6 \times 3^x - 27 = 0$

3

- c) There are five nominees for President and Vice President of a club. Three are women and two are men. The first name, selected at random, will be the President and the second name will be the Vice President.

4

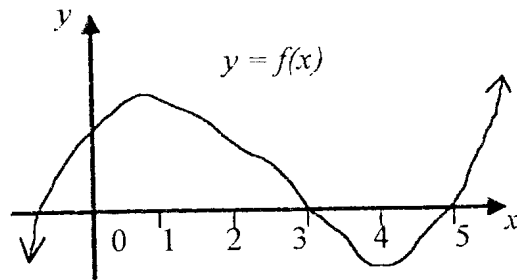
- i) Draw a tree diagram to represent all possible outcomes
- ii) Determine the probability that the two positions will be filled by a woman and a man, in either order.

QUESTION 6 Use a SEPARATE Writing Booklet.**Marks**

- a) Given the graph of $y = f(x)$, EXPLAIN why

3

$$\int_0^4 f(x)dx \text{ is LESS than } \int_0^3 f(x)dx$$



- b) A sector of a circle, of radius 1 cm, has a perimeter of 4 cm.

4

- i) Show that the angle at the centre of the sector is 2 radians
- ii) Find the area of the sector

- c) Use Simpson's Rule with 5 function values (i.e. 4 strips) to find an approximation for $\int_1^5 (\log_e x)dx$ correct to 3 decimal places

5

QUESTION 7 Use a SEPARATE Writing Booklet.

Marks

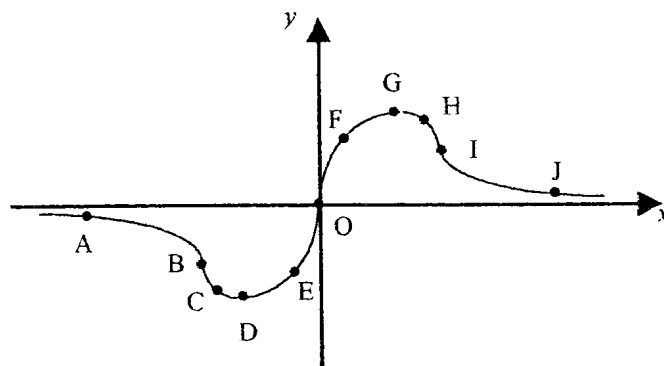
- a) The rate of decay of a radioactive substance is proportional to the mass, M , present at time, t years, i.e. $\frac{dM}{dt} = -kM$

5

- i) Show that $M = M_0 e^{-kt}$ satisfies $\frac{dM}{dt} = -kM$
- ii) If the half-life of the substance is 17 600 years, find k (correct to 6 decimal places)
- iii) How long will it take for $\frac{2}{3}$ (two thirds) of the substance to decay

- b) For the given graph of $y = f(x)$ write down which of the labelled point(s) best demonstrate the properties below:

7



- i) $f(x) = 0$
- ii) $f'(x) = 0$
- iii) $f''(x) = 0$
- iv) $f(x) > 0$
- v) $f'(x) > 0$
- vi) $f''(x) > 0$
- vii) $\lim_{x \rightarrow \infty} f(x) = 0$

QUESTION 8 Use a SEPARATE Writing Booklet.**Marks**

- a) i) Differentiate $y = \cos^3 x$ **5**
- ii) Hence, evaluate $\int_0^{\frac{\pi}{4}} (\cos^2 x \sin x) dx$
- b) Consider the parabola with the equation $x^2 - 8x = 12y - 28$ **5**
- i) Show that the equation can be written as $(x - 4)^2 = 12(y - 1)$
- ii) Find the coordinates of the vertex
- iii) Find the coordinates of the focus
- iv) Find the equation of the directrix
- c) Find k if $\int_1^k \left(\frac{1}{x}\right) dx = 1$ **2**

QUESTION 9 Use a SEPARATE Writing Booklet.

A particle moves along the x -axis so that its displacement, x metres, after t seconds is given by $x = 3 - 2 \cos t$ **12**

- i) Find the initial displacement
- ii) Show that the particle starts from rest
- iii) When does the particle next come to rest?
- iv) Find the velocity when the particle passes through $x = 2$ for the second time
- v) Find the particle's greatest velocity
- vi) Find the particle's position when it is NOT being accelerated

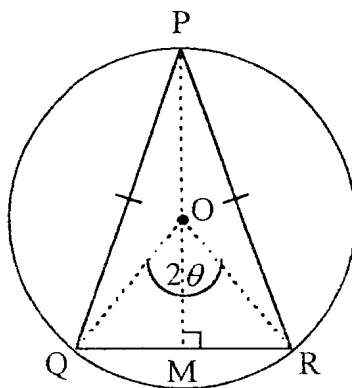
QUESTION 10 Use a SEPARATE Writing Booklet.

Marks

a) i) Show that $\frac{1}{x^2-9} = \frac{1}{6} \left(\frac{1}{x-3} - \frac{1}{x+3} \right)$ 5

ii) Hence find the exact volume generated by revolving $y = \frac{1}{\sqrt{x^2-9}}$ around the x -axis from $x=5$ to $x=6$

- b) Isosceles triangle PQR is in a circle of radius 1 unit, centre O. 7
 $\angle QOR = 2\theta$ (θ is acute). PO is extended to meet QR at M such that $\angle OMR = 90^\circ$



- i) Prove that $QM = \sin \theta$ and $OM = \cos \theta$
- ii) Show that the area, A , of $\triangle PQR$ is given by $A = \sin \theta (\cos \theta + 1)$
- iii) Hence show that $\triangle PQR$ has a maximum area when it is equilateral

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

⑤ continued:

$$(b) \quad 9^x + 6 \times 3^x - 27 = 0$$

$$9^x = (3^2)^x = (3^x)^2$$

$\therefore$ Letting $A = 3^x$ gives: $\frac{1}{2}$

$$A^2 + 6A - 27 = 0$$

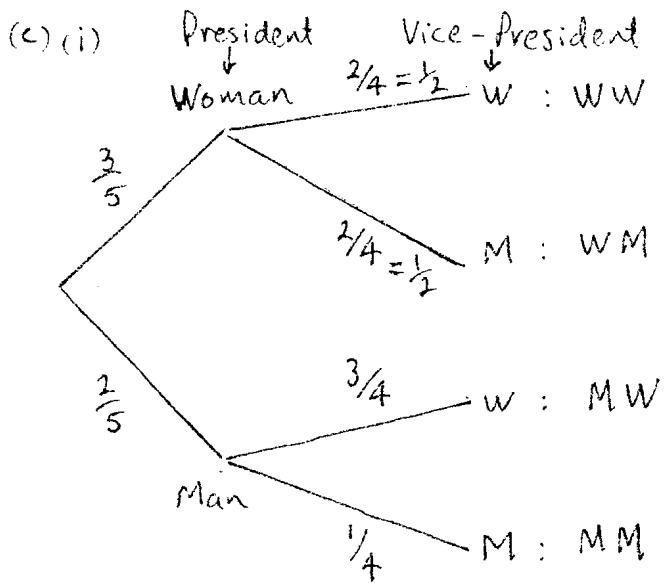
$$(A+9)(A-3) = 0$$

$$\therefore A = -9 \text{ or } A = 3$$

i.e. $3^x = -9^{\frac{1}{2}}$ $3^x = 3^{\frac{1}{2}}$

no solution $\frac{1}{2}$ $3^x = 3^{\frac{1}{2}}$

$$\therefore x = 1$$



Note President can not be Vice-Pres.

∴ names selected but not replaced

(ii) $p(WM \text{ or } MW)$

$$= \frac{3}{5} \times \frac{1}{2} + \frac{2}{5} \times \frac{3}{4}$$

$$= \frac{3}{10} + \frac{3}{10}$$

$$= \left(\frac{3}{5} \right) \text{ (or } 60\%)$$

⑥ (a) $\int_0^3 f(x) dx$ is positive, say $+q$: (1)

$\int_3^4 f(x) dx$ is negative, say -2

(NOTE Area from 0 to 3 > Area from 3 to 4)

or $|+9| > |-2|$)

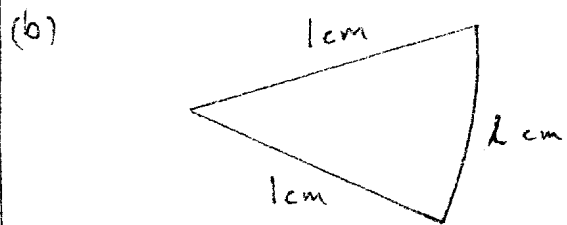
$$\text{Now: } \int_0^4 f(x) dx = \int_0^3 f(x) dx + \int_3^4 f(x) dx$$

$$\left. \begin{aligned} &= 9 + (-2) \\ &= +7 \dots\dots (2) \end{aligned} \right\} \frac{1}{2}$$

$\therefore$ Comparing (1) and (2):

$7 < 9$

$$\text{i.e. } \int_0^4 f(x) dx < \int_0^3 f(x) dx \quad (\text{QED})$$



(i) $2+1+1 = 4\text{ cm}$ (perimeter) $\frac{1}{2}$

$$\therefore L+2=2$$

$$L = 2 \text{ cm}$$

$$l = r\omega \therefore \omega = l/r$$

$$= 2/1$$

i.e. $\theta = 2 \text{ rad.}$ QED $\frac{1}{2}$

(ii) $A = \frac{1}{2} v^2 \odot$

$$= \frac{1}{2} \times 1^2 \times 2$$

$$\therefore A = 1 \text{ cm}^2$$

1) (a) on calculator:

$$\sqrt{(4\pi) \div (3.6x^2 - 9.8)} = 1.1218...$$

4 sig. fig's

$\therefore$ Answer = 1.122

(b) $\frac{6}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} = \frac{6(\sqrt{3}+1)}{(\sqrt{3})^2 - (1)^2}$

$$= \frac{6(\sqrt{3}+1)}{3-1}$$

$$= \frac{3 \cancel{6} (\sqrt{3}+1)}{\cancel{2}}$$

$$= \underline{3(\sqrt{3}+1)}$$

(c) $d/dx (6-x^3) = 0 - 3x^2$

$$= \underline{-3x^2}$$

d) $\frac{x \times 3}{2} + \frac{x \times 2}{3} = 1$

$$\therefore 3x + 2x = 6$$

$$\therefore 5x = 6$$

$$\underline{x = 6/5 \text{ or } 1\frac{1}{5} \text{ or } 1.2}$$

2) $\int 4/x \, dx = 4 \int 1/x \, dx$

$$= \underline{4 \log_e x + C}$$

3) $9 - 16t^2 = (3)^2 - (4t)^2$

$$\{a^2 - b^2 = (a+b)(a-b)\}$$

$$= \underline{(3+4t)(3-4t)}$$

2) (a) (i) $y = e^{\sin x} + \frac{1}{2}x^4$

$$\therefore \frac{dy}{dx} = \cos x \times e^{\sin x} + \frac{1}{2} \times 4x^3$$

$$= \underline{\cos x \cdot e^{\sin x} + 2x^3}$$

(ii) $y = \frac{\log_e x}{x} \rightarrow u = \log_e x, v = x$
 $u' = \frac{1}{x}, v' = 1$

$$\therefore y' = \frac{vu' - uv'}{v^2}$$

$$= \frac{x \times \frac{1}{x} - \log_e x \times 1}{x^2}$$

$$= \underline{\frac{1 - \log_e x}{x^2}}$$

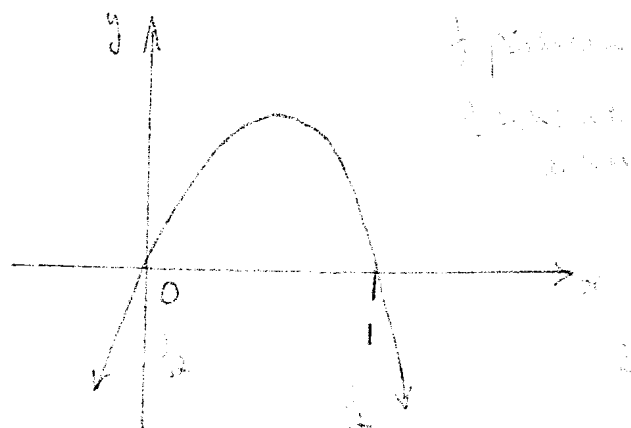
(b) $y = -x^2 + x + 0$ $\leftarrow$ y-intercept

and for $x - x^2 = 0$

$$x(1-x) = 0$$

$$\therefore x = 0 \text{ or } 1 \leftarrow x\text{-intercepts}$$

and parabola; concave down



② continued:

$$(c) |x+4|=1$$

$$\therefore x+4=1 \text{ or } -(x+4)=1$$

$$\therefore x+4=-1$$

$$\therefore x = -3 \text{ or } \therefore x = -5$$

$$(d) \sec 210^\circ = \frac{1}{\cos 210^\circ}$$

and for $\cos 210^\circ$:

$$\text{3rd quad: } \cos(180^\circ + 30^\circ)$$

$$= -\cos 30^\circ$$

$$= -\frac{\sqrt{3}}{2}$$

$$\therefore \sec 210^\circ = \frac{1}{-\frac{\sqrt{3}}{2}}$$

$$\left\{ \begin{array}{l} \text{Answer can be} \\ \text{checked on} \\ \text{calculator} \end{array} \right\} = \frac{-2}{\sqrt{3}}$$

③(a) Arithmetic, $a=6$, $d=9$.

$$(i) T_n = a + (n-1)d$$

$$= 6 + (n-1) \times 9$$

$$= 6 + 9n - 9$$

$$\text{i.e. } T_n = 9n - 3$$

$$(ii) \text{ Let } 9n - 3 = 4623$$

$$9n = 4626$$

$$n = 514$$

$$\text{i.e. } 514^{\text{th}} \text{ term}$$

$$(b) (i) \text{ Gradient}_{AB} = \frac{-6-3}{11-(-1)} \left(\frac{y_2-y_1}{x_2-x_1} \right) \\ = \frac{-9}{12} \\ = \left(-\frac{3}{4} \right)$$

$$(ii) y - y_1 = m(x - x_1) \\ y - 3 = -\frac{3}{4}(x + 1)$$

$$\therefore 4y - 12 = -3x - 3$$

$$\therefore 3x + 4y - 9 = 0 \quad (\text{QED!})$$

(or: can be done by showing - by substitution - that A and B satisfy equation). (or 1, 1)

$$(iii) \parallel \text{ to } AB \therefore m_L = -\frac{3}{4}$$

$$\text{and through origin: } y = mx + b$$

$$\text{i.e. } y = mx$$

$$\therefore y = -\frac{3}{4}x$$

$$(iv) (4, k) \text{ on } L \therefore k = -\frac{3}{4} \times 4$$

$$k = -3$$

$$(v) d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|3 \times 4 + 4 \times -3 + -9|}{\sqrt{3^2 + 4^2}}$$

$$= \frac{|12 - 12 - 9|}{\sqrt{25}}$$

$$= \frac{|-9|}{5} = \frac{9}{5}$$

(3)

④ (a) (i) $\int \cos ax \, dx = \frac{1}{a} \sin ax + c$
(from Standard Integrals Sheet)

$\therefore \int \cos 2x \, dx = \frac{1}{2} \sin 2x + c$

(ii) $\int \frac{dx}{2x+3} = \frac{1}{2} \int \frac{2}{2x+3} \, dx$
 $= \frac{1}{2} \log_e (2x+3) + c$

(iii) $\int e^{3x} \, dx = \frac{1}{3} \int 3e^{3x} \, dx$
 $= \frac{1}{3} e^{3x} + c$

(b) X: $A = P(1 + r/100)^n$
 $= 5000 (1 + 9/100)^6$
 $= 5000 \times 1.09^6$
 $= \$8385.50 \text{ (rounded)}$
 $\therefore \text{C.I.} = 8385.50 - 5000$
 $= \$3385.50 \dots \dots (1)$

Y: $SI = \frac{Pnr}{100}$
 $= \frac{5000 \times 9 \times 6}{100}$
 $= \$2700 \dots \dots (2)$

$\therefore \text{Difference} = (1) - (2)$
 $= 3385.50 - 2700$
 $= \$685.50$

(c) Pos. definite if:

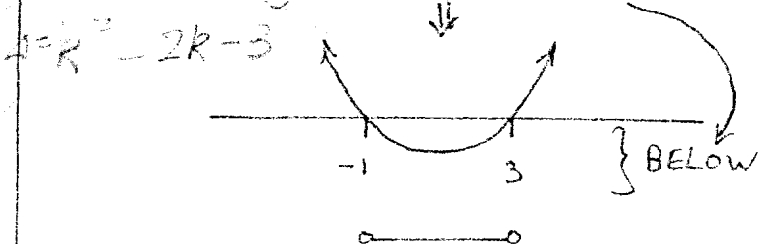
$a > 0$ AND $\Delta < 0$

ie. $3-k > 0$ $b^2 - 4ac < 0$

$3 > k$ $(3-k)^2 - 4(3-k) < 0$

or $k < 3$ $(3-k)[(3-k)-4] < 0$

$\dots (1)$ $(3-k)(-1-k) < 0$



ie. $-1 < k < 3 \dots \dots (2)$

from (1) AND (2) we have:

$-1 < k < 3$

⑤ (a) (i) $\sin \alpha = \frac{AB}{AD}$ (from $\triangle ABD$)
 $= \frac{1x}{2x}$
 $= \frac{1}{2}$

$\therefore \alpha = \sin^{-1}(\frac{1}{2})$

$\alpha = 30^\circ$ (Q.E.D.)

(ii) $\angle ADC = 30^\circ$ (above)

$\therefore \angle BAD = 60^\circ$ ($\angle$ sum of $\triangle ABD$)

$\therefore \angle DAC = 30^\circ$

(iii) $\triangle ACD$ is isosceles

$\therefore AC = DC = 2$

$\therefore$ In $\triangle ABC$:

$\cos \alpha = \frac{AB}{2}$

ie. $\cos 30^\circ = \frac{AB}{2} = \frac{\sqrt{3}}{2}$

$\therefore AB = \sqrt{3} \text{ cm}$

⑥ continued:

(c) $h = \frac{b-a}{n} = \frac{5-1}{4} = 1$ (strip width) $\frac{1}{2}$ 5 function values

$$= \frac{5-1}{4}$$

$$= 1 \text{ (strip width) } \frac{1}{2}$$

x	$y (\log_e x = \ln x)$	x	$=$
1	$\ln 1 = 0$	1	0
2	$\ln 2 = 0.6931$	4	2.7724
3	$\ln 3 = 1.0986$	2	2.1972
4	$\ln 4 = 1.3863$	4	5.5452
5	$\ln 5 = 1.6094$	1	1.6094
TOTAL = 12.1242			

$$\therefore \int_1^5 \log_e x \, dx \approx \frac{h}{3} \times \text{TOTAL}$$

$$= \frac{1}{3} \times 12.1242$$

$$= 4.0414$$

$$= \boxed{4.041} \text{ (3dp)}$$

(NOTE: Graph of $y = \log_e x$ above

x -axis for $x=1$ to $x=5$

$$\therefore \int_1^5 \log_e x \, dx = \text{Area})$$

⑦(a)(i) $M = M_0 e^{-kt}$

$$\therefore \frac{dM}{dt} = M_0 \cdot -ke^{-kt}$$

$$= -k(M_0 e^{-kt})$$

$$\text{i.e. } \frac{dM}{dt} = -kM \text{ (QED)}$$

(ii) $M = \frac{1}{2} M_0$ at $t = 17600$

$$\text{i.e. } \frac{1}{2} M_0 = M_0 e^{-17600k}$$

$$\therefore \frac{1}{2} = e^{-17600k}$$

$$\ln \frac{1}{2} = -17600k$$

$$\therefore k = \ln \frac{1}{2} \div -17600$$

$$\text{i.e. } \boxed{k = 0.000039} \text{ (6dp)}$$

(iii) For $\frac{2}{3}$ decayed, $M = \frac{1}{3} M_0$

$$\text{i.e. } \frac{1}{3} = e^{-0.000039t}$$

$$\ln \frac{1}{3} = -0.000039t$$

$$\therefore t = \ln \frac{1}{3} \div -0.000039$$

$$= 28169.5 \dots$$

$$\text{i.e. } \boxed{28200 \text{ years}} \text{ (3 sig. figs)}$$

b)(i) $f(x) = 0 \rightarrow x$ -intercept(s) : $\textcircled{O}$

(ii) $f'(x) = 0 \rightarrow$ st. points : $\textcircled{D, G}$

(iii) $f''(x) = 0 \rightarrow$ inf. pts : $\textcircled{B, O, I}$

(iv) $f(x) > 0 \rightarrow$ above x -axis : $\textcircled{F, G, H, I, J}$

(v) $f'(x) > 0 \rightarrow$ increasing : $\textcircled{E, O, F}$

(vi) $f''(x) > 0 \rightarrow$ concave up : $\textcircled{C, D, E, J}$

(v) $\lim_{x \rightarrow \infty} f(x) = 0 \rightarrow$ approaches x -axis as $x \uparrow$ in positive direction : $\textcircled{J}$

correct $\left(\frac{1}{2}\right)$

incorrect $\left(-\frac{1}{2}\right)$

8 (a)(i) $y = \cos^3 x$

$\therefore y = (\cos x)^3$

$\therefore \frac{dy}{dx} = 3(\cos x)^2 \times -\sin x$
 $= -3 \cos^2 x \sin x$

(ii) $\int_0^{\pi/4} (\cos^2 x \sin x) dx$
 $= -\frac{1}{3} \int_0^{\pi/4} (-3 \cos^2 x \sin x) dx$
 $= -\frac{1}{3} [\cos^3 x]_0^{\pi/4}$ (from (i))
 $= -\frac{1}{3} [(\cos \pi/4)^3 - (\cos 0)^3]$
 $= -\frac{1}{3} [(\frac{1}{\sqrt{2}})^3 - (1)^3]$
 $= -\frac{1}{3} [0.3536 - 1]$ (4dp)
 $= 0.215$ 3dp

or $-\frac{1}{3} (\frac{1}{2\sqrt{2}} - 1) = \frac{1}{3} (1 - \frac{1}{2\sqrt{2}})$

b)(i) $x^2 - 8x + 16 = 12y - 28 + 16$

$(x-4)^2 = 12y - 12$

$\therefore (x-4)^2 = 12(y-1)$ can also be done by removing brackets and rearranging
 ie. $(x-4)^2 = 12(y-1)$ (QED)

Using: $(x-h)^2 = 4a(y-k)$

Vertex at (h, k) , focal length = a

$\therefore$ (ii) Vertex $(4, 1)$

(iii) Concave up $\therefore$ Focus at

$(4, 1+a)$

where $4a = 12 \therefore a = 3$

$\therefore$ focus at $(4, 1+3)$
 $(4, 4)$

(iv) Directrix:

$y = 1 - a$
 $= 1 - 3$

ie. $y = -2$

(c) $\int_1^k \frac{1}{x} dx = 1$

$\therefore [\ln x]_1^k = 1$

$\therefore \ln k - \ln 1 = 1$

$\therefore \ln k - 0 = 1$

$\ln k = 1$

or $\log_e k = 1$

$\therefore e^1 = k$

ie. $k = e$

9 $x = 3 - 2 \cos t$

(i) at $t = 0$, $x = 3 - 2 \cos 0$

$= 3 - 2 \times 1$

$= 3 - 2$

$\therefore x = 1$ m

(ii) $x = 3 - 2 \cos t$

$\therefore v = \dot{x} = 0 - 2 \times (-\sin t)$

$= 2 \sin t$

$\therefore$ at $t = 0$, $v = 2 \sin 0$

$= 2 \times 0$

$v = 0 \rightarrow$ stationary

7

1

4

1

1

2

$$\left. \begin{array}{l} 60^\circ \\ \text{or} \\ \pi/3 \end{array} \right)$$

2

+ 1 revolution

4

•

1

7

2

○

1.

10

} "graphical"
approach

1

1

4

1

10 (a) (i) Show: $\frac{1}{x^2-9} = \frac{1}{6} \left(\frac{1}{x-3} - \frac{1}{x+3} \right)$

$$\text{RHS} = \frac{1}{6} \left(\frac{1 \times x+3}{x-3 \times x+3} - \frac{1 \times x-3}{x+3 \times x-3} \right)$$

$$= \frac{1}{6} \left(\frac{x+3 - (x-3)}{x^2-9} \right)$$

$$= \frac{1}{6} \left(\frac{x+3-x+3}{x^2-9} \right)$$

$$= \frac{1}{6} \left(\frac{6}{x^2-9} \right)$$

$$= \frac{1}{x^2-9}$$

= LHS (QED)

(ii) $V = \pi \int_5^6 \left(\frac{1}{\sqrt{x^2-9}} \right)^2 dx$

$$= \pi \int_5^6 \frac{1}{x^2-9} dx$$

$$= \pi \int_5^6 \frac{1}{6} \left(\frac{1}{x-3} - \frac{1}{x+3} \right) dx \quad (\text{from (i)})$$

$$= \frac{\pi}{6} \int_5^6 \frac{1}{x-3} - \frac{1}{x+3} dx$$

$$= \frac{\pi}{6} \left[\ln(x-3) - \ln(x+3) \right]_5^6$$

$$= \frac{\pi}{6} \left[\ln \left(\frac{x-3}{x+3} \right) \right]_5^6$$

$$= \frac{\pi}{6} \left[\ln \left(\frac{6-3}{6+3} \right) - \ln \left(\frac{5-3}{5+3} \right) \right]$$

$$= \frac{\pi}{6} \left[\ln \frac{1}{3} - \ln \frac{1}{4} \right]$$

$$= \frac{\pi}{6} \ln \left(\frac{1/3}{1/4} \right)$$

$$= \frac{\pi}{6} \ln \frac{4}{3} \text{ units}^3 \quad (= 0.15 \text{ 2dp})$$

(b) (i) $\angle MOQ = \frac{1}{2} \times \angle ROQ$
 $= \theta$

In ΔMOQ : $\sin \theta = \frac{QM}{OQ}$
 ie $\sin \theta = \frac{QM}{1}$ } $OQ = \text{radius} = 1$

$\therefore QM = \sin \theta$ (QED)

Similarly: $\cos \theta = \frac{OM}{OQ}$

$\therefore OM = \cos \theta$ (QED)

(ii) Area $\Delta PQR = \frac{1}{2}bh$

$$= \frac{1}{2} \times QR \times MP$$

but: $\frac{1}{2} \times QR = QM = \sin \theta$ (1)

and: $MP = OM + OP$

$$= \cos \theta + \text{radius} \approx 1$$

$$= \cos \theta + 1$$
 (2)

$\therefore$ from (1)/(2): Area = $\sin \theta (\cos \theta + 1)$ (QED)

(iii) Area max when $\frac{dA}{d\theta} = 0$ ($= A'$)

using product rule: $u = \sin \theta$, $v = (\cos \theta + 1)$
 $u' = \cos \theta$ $v' = -\sin \theta$

$$\therefore A' = (\cos \theta + 1) \cos \theta + \sin \theta \times (-\sin \theta)$$

$$= \cos^2 \theta + \cos \theta - \sin^2 \theta$$

$$= \cos^2 \theta + \cos \theta - (1 - \cos^2 \theta)$$

$$= 2\cos^2 \theta + \cos \theta - 1$$

$\therefore$ for: $2\cos^2 \theta + \cos \theta - 1 = 0$

$$(2\cos \theta - 1)(\cos \theta + 1) = 0$$

$\therefore \cos \theta = \frac{1}{2}$ or $\cos \theta = -1$

$\therefore \theta = 60^\circ$ (or 270° : not possible)

and if $\theta = 60^\circ$, $\angle ROQ = 120^\circ$

$\therefore \angle PQR = \angle PRQ = 60^\circ$ $\therefore$ equilateral

AND:

θ	59°	60°	61°
A'	+0.05	0	-0.05

} easier than A'' test?

$\Rightarrow$ $\therefore A_{\text{max}}$ (when $\theta = 60^\circ$)